

## PANJGUR, Pakistan

WMO: 417390

Lat: 26.955N

Lon: 64.133E

Elev: 3289

StdP: 13.03

Time Zone: 5.00 (E05)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 33.7       | 36.3 | 9.9                           | 10.3 | 59.3 | 15.0 | 13.3 | 56.8 | 26.8                  | 71.1 | 22.6 | 64.1 | 5.9                      | 0    | 0.429 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 23.2                   | 106.6           | 70.7 | 104.5 | 70.3 | 102.6 | 70.0 | 77.9                | 96.8 | 76.3 | 96.0 | 74.9 | 94.6 | 10.5                 | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.1                            | 139.1 | 88.2 | 70.9 | 128.8 | 87.3 | 69.2 | 121.5 | 88.0 | 44.1          | 97.4 | 42.4 | 95.6 | 40.9 | 94.8 | 90.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 23.4              | 19.7 | 16.7 | DB                         | 29.8 | 110.3              | 2.8 | 2.2                                                | 27.8 | 111.9      | 26.2 | 113.2      | 24.6 | 114.4      | 22.6 | 116.0 |  |
|                   |      |      | WB                         | 25.1 | 79.7               | 3.2 | 1.6                                                | 22.8 | 80.8       | 20.9 | 81.8       | 19.1 | 82.7       | 16.8 | 83.9  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 74.4   | 53.1  | 57.1  | 67.5  | 78.9  | 86.8  | 91.0  | 91.4  | 88.1  | 83.2  | 75.7  | 63.9  | 54.4 |
|                                                                               | DBStd       | 14.86  | 5.56  | 7.77  | 7.12  | 5.54  | 4.47  | 3.77  | 2.79  | 2.96  | 3.65  | 4.69  | 6.39  | 5.96 |
|                                                                               | HDD50       | 91     | 34    | 27    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 29   |
|                                                                               | HDD65       | 1087   | 369   | 235   | 57    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 96    | 328  |
|                                                                               | CDD50       | 8981   | 131   | 224   | 543   | 868   | 1142  | 1231  | 1285  | 1180  | 997   | 796   | 418   | 166  |
|                                                                               | CDD65       | 4501   | 0     | 13    | 135   | 420   | 677   | 781   | 820   | 715   | 547   | 331   | 62    | 0    |
|                                                                               | CDH74       | 65293  | 11    | 201   | 1497  | 5241  | 10155 | 12510 | 12946 | 10668 | 7424  | 3918  | 689   | 33   |
|                                                                               | CDH80       | 39355  | 0     | 24    | 520   | 2733  | 6367  | 8366  | 8570  | 6621  | 4222  | 1792  | 139   | 1    |
| Wind                                                                          | WSAvg       | 6.4    | 6.3   | 6.6   | 6.4   | 6.2   | 6.3   | 8.1   | 7.0   | 6.1   | 5.2   | 5.6   | 6.1   | 6.5  |
| Precipitation                                                                 | PrecAvg     | 3.5    | 0.6   | 0.6   | 0.6   | 0.3   | 0.1   | 0.1   | 0.6   | 0.3   | 0.1   | 0.0   | 0.1   | 0.4  |
|                                                                               | PrecMax     | 7.9    | 2.9   | 3.8   | 2.7   | 2.6   | 0.9   | 1.0   | 4.5   | 2.1   | 1.0   | 1.7   | 1.7   | 3.3  |
|                                                                               | PrecMin     | 0.4    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                               | PrecStd     | 1.9    | 0.7   | 0.8   | 0.7   | 0.5   | 0.2   | 0.2   | 1.0   | 0.6   | 0.2   | 0.2   | 0.2   | 0.7  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 75.3  | 83.2  | 93.0  | 100.4 | 107.3 | 108.9 | 108.7 | 107.0 | 101.2 | 97.3  | 87.0  | 77.5 |
|                                                                               |             | MCWB   | 55.3  | 58.0  | 64.4  | 67.3  | 71.5  | 70.1  | 71.7  | 70.6  | 67.5  | 65.7  | 58.9  | 55.3 |
|                                                                               | 2%          | DB     | 72.0  | 79.5  | 89.3  | 97.2  | 104.2 | 106.7 | 106.0 | 103.3 | 98.8  | 94.3  | 83.9  | 74.0 |
|                                                                               |             | MCWB   | 53.4  | 56.8  | 62.9  | 66.0  | 70.3  | 70.6  | 70.6  | 69.5  | 67.1  | 64.5  | 58.8  | 53.7 |
|                                                                               | 5%          | DB     | 69.5  | 76.3  | 86.0  | 95.0  | 101.9 | 104.4 | 104.1 | 101.0 | 97.2  | 91.7  | 81.1  | 71.3 |
|                                                                               |             | MCWB   | 52.0  | 55.8  | 61.7  | 65.3  | 70.8  | 70.3  | 70.4  | 68.4  | 66.6  | 63.7  | 58.2  | 52.7 |
|                                                                               | 10%         | DB     | 66.8  | 72.7  | 82.3  | 92.4  | 99.5  | 102.3 | 102.0 | 99.1  | 95.5  | 88.9  | 78.1  | 68.2 |
|                                                                               |             | MCWB   | 50.4  | 54.0  | 60.1  | 64.2  | 69.5  | 69.6  | 70.2  | 68.0  | 66.1  | 62.3  | 57.2  | 50.9 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 58.5  | 61.7  | 68.9  | 70.8  | 81.9  | 78.5  | 78.9  | 79.8  | 76.2  | 71.2  | 63.8  | 59.4 |
|                                                                               |             | MCDB   | 70.7  | 78.8  | 89.3  | 94.2  | 97.6  | 100.4 | 97.3  | 93.8  | 94.4  | 90.0  | 80.0  | 70.4 |
|                                                                               | 2%          | WB     | 56.1  | 58.6  | 65.4  | 68.8  | 77.1  | 76.8  | 77.2  | 76.7  | 73.5  | 68.2  | 61.6  | 56.3 |
|                                                                               |             | MCDB   | 69.0  | 75.5  | 84.8  | 92.2  | 97.8  | 97.9  | 96.4  | 93.1  | 93.1  | 89.2  | 78.4  | 70.1 |
|                                                                               | 5%          | WB     | 54.0  | 56.7  | 63.1  | 67.4  | 74.3  | 75.2  | 75.9  | 74.1  | 71.3  | 66.3  | 60.0  | 54.4 |
|                                                                               |             | MCDB   | 67.2  | 73.3  | 82.1  | 90.6  | 96.9  | 97.2  | 95.2  | 91.0  | 90.7  | 87.5  | 77.6  | 68.5 |
|                                                                               | 10%         | WB     | 51.6  | 54.8  | 61.2  | 65.9  | 71.8  | 73.4  | 74.7  | 72.5  | 69.5  | 64.4  | 58.1  | 52.2 |
|                                                                               |             | MCDB   | 64.4  | 70.6  | 79.4  | 88.5  | 95.2  | 94.9  | 93.8  | 90.2  | 89.2  | 85.4  | 76.0  | 66.2 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 24.9  | 24.4  | 24.8  | 25.0  | 25.2  | 23.2  | 22.9  | 23.7  | 24.6  | 26.1  | 24.6  | 25.1 |
|                                                                               |             | MCDBR  | 28.7  | 28.1  | 28.5  | 27.6  | 27.0  | 25.4  | 25.4  | 25.6  | 26.3  | 28.3  | 27.3  | 28.5 |
|                                                                               |             | MCWBR  | 15.9  | 14.3  | 12.3  | 10.2  | 10.5  | 8.6   | 8.4   | 8.5   | 9.7   | 11.1  | 12.9  | 15.6 |
|                                                                               | 5% WB       | MCDBR  | 26.0  | 25.3  | 26.9  | 26.1  | 25.8  | 23.2  | 22.8  | 23.3  | 24.9  | 27.3  | 25.4  | 26.5 |
|                                                                               |             | MCWBR  | 16.1  | 14.9  | 13.4  | 11.0  | 13.4  | 10.0  | 9.0   | 9.6   | 10.8  | 12.7  | 13.4  | 16.2 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.336  | 0.384 | 0.433 | 0.499 | 0.534 | 0.562 | 0.632 | 0.560 | 0.460 | 0.398 | 0.350 | 0.327 |      |
|                                                                               | taud        | 2.332  | 2.178 | 2.011 | 1.867 | 1.796 | 1.750 | 1.654 | 1.774 | 1.946 | 2.098 | 2.279 | 2.353 |      |
|                                                                               | Ebn at Noon | 288    | 280   | 272   | 257   | 247   | 239   | 223   | 238   | 260   | 271   | 280   | 286   |      |
|                                                                               | Edh at Noon | 35     | 44    | 55    | 65    | 70    | 73    | 80    | 70    | 58    | 47    | 36    | 33    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1329   | 1576  | 1930  | 2195  | 2354  | 2398  | 2223  | 2172  | 2065  | 1806  | 1485  | 1281  |      |
|                                                                               | RadStd      | 80     | 96    | 80    | 83    | 53    | 88    | 104   | 83    | 55    | 42    | 41    | 59    |      |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                       | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page